

A CUSTOMISED Li-Fi LINK For Seamless Communication

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Light fidelity (Li-Fi) is a widely adopted form of wireless communication that uses optical energy to verify device functionality.

A key challenge lies in establishing communication between two heterogeneous devices over extended distances, especially under full daylight conditions.

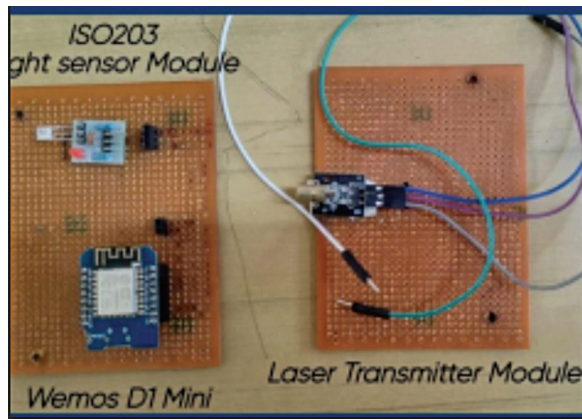


Fig. 1: Authors' prototype setup

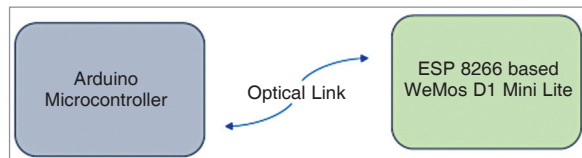


Fig. 2: Block diagram representation of the setup

to 1ms leads to errors and the loss of alphanumeric characters.

The system has been validated to sustain reliable communication across a 152cm (60-inch) distance between the transmitter (Tx) and receiver (Rx) modules, even in bright lighting.

The setup confirms the viability of using different microcontrollers—for instance, an ESP8266-based WeMos D1 Mini at the transmitter end and another ESP8266 module at the receiver end. To introduce heterogeneity, other combinations such as ESP8266 with Arduino, ESP32

TABLE 1 BILL OF MATERIALS		
Components	Description	Quantity
WeMos D1 Mini	For programming of Rx	1
WeMos D1 Mini	For programming of Tx	1
ISO 203 Rx Module	To receive the data bit sequence	1
Laser Tx module	To transmit the data bit sequence	1
Jumper wires	Male-female jumper wires to connect	As required

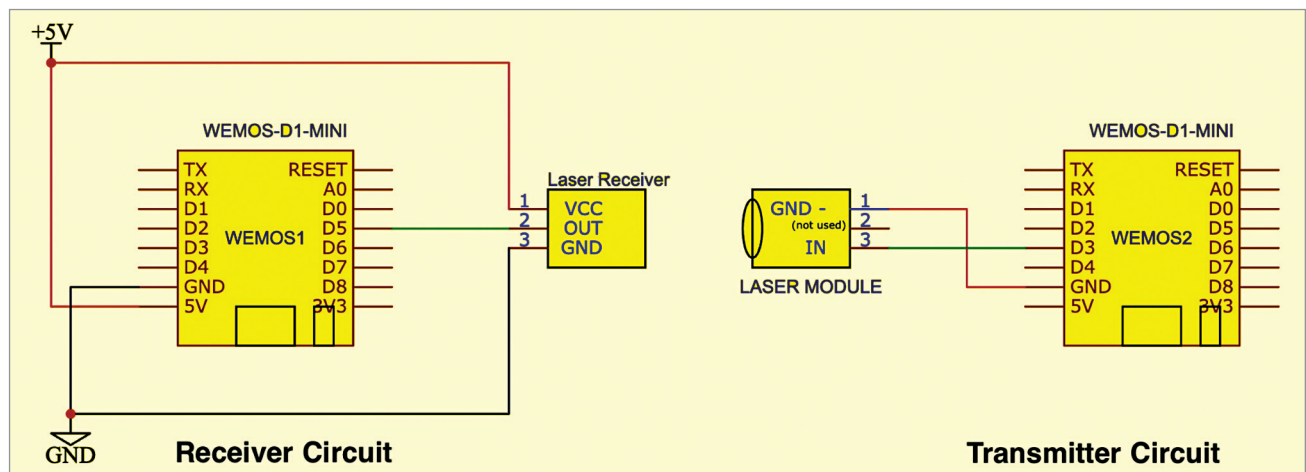


Fig. 3: Combined circuit diagram of transmitter and receiver